



Peptic ulcer disease

Surgical treatment of PUD

Surgery is now rarely required for peptic ulcer disease but it is needed in some cases.

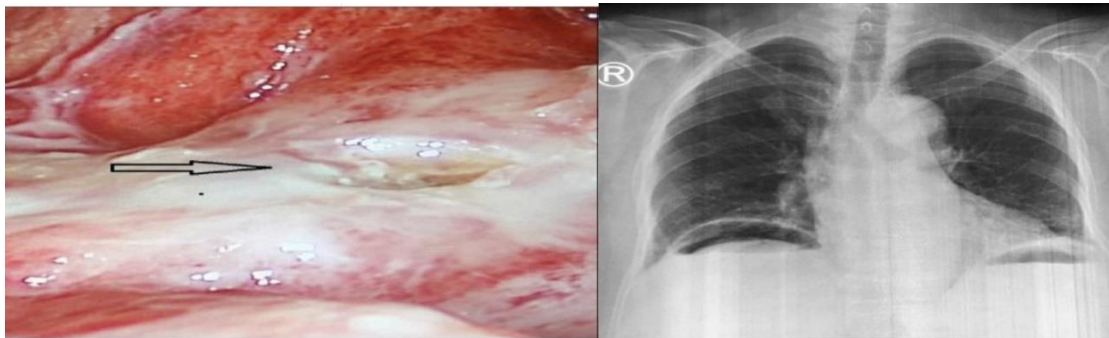
i	21.37 Indications for surgery in peptic ulcer
Emergency	
• Perforation • Haemorrhage	
Elective	
• Gastric outflow obstruction • Persistent ulceration despite adequate medical therapy • Recurrent ulcer following gastric surgery	

Complications of peptic ulcer disease: Perforation

- When perforation occurs, the contents of the stomach escape into the peritoneal cavity, leading to peritonitis. This is more common in duodenal than in gastric ulcers and is usually found with ulcers on the anterior wall .
- About one-quarter of all perforations occur in acute ulcers and NSAIDs are often incriminated .
- Perforation can be the first sign of ulcer and a history of recurrent epigastric pain is uncommon .
- The most striking symptom is sudden, severe pain; its distribution follows the spread of the gastric contents over the peritoneum.
- The pain initially develops in the upper abdomen and rapidly becomes generalised; shoulder tip pain is caused by irritation of the diaphragm. The pain is accompanied by shallow respiration, due to limitation of diaphragmatic movements, and by shock. The abdomen is held immobile and there is generalised 'board-like' rigidity. Bowel sounds are absent and liver dullness to percussion decreases due to the presence of gas under the diaphragm .
- After some hours, symptoms may improve, although abdominal rigidity remains. Later, the patient's condition deteriorates as general peritonitis develops. In at least 50% of cases, an erect chest X-ray shows free air beneath the diaphragm.

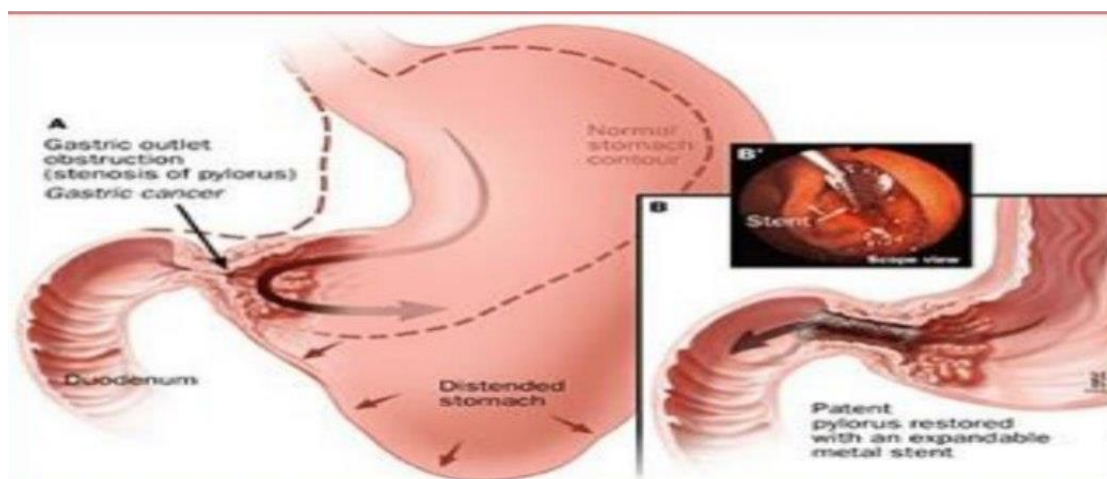


- If not, a water-soluble contrast swallow will confirm leakage of gastroduodenal contents .
- After resuscitation, the acute perforation should be treated surgically, either by simple closure or by conversion of the perforation into a pyloroplasty if it is large.
- Following surgery, H. pylori should be treated (if present) and NSAIDs avoided. Perforation carries a mortality of 25%, reflecting the advanced age and significant comorbidity of the population that are affected.



Complications of peptic ulcer disease: Gastric outlet obstruction

- The most common is an ulcer in the region of the pylorus. The presentation is with nausea , vomiting and abdominal distension. Large quantities of gastric content are often vomited and food eaten 24 hours or more previously may be recognized .
- Physical examination may show evidence of wasting and dehydration. A succussion splash may be elicited 4 hours or more after the last meal or drink.





- Visible gastric peristalsis is diagnostic of gastric outlet obstruction.
- Loss of acidic gastric contents leads to alkalosis and dehydration with low serum chloride and potassium and raised serum bicarbonate and urea concentrations (hypochloraemic metabolic alkalosis). Paradoxical aciduria occurs because of enhanced renal absorption of Na^+ in exchange for H^+ .
- Endoscopy should be performed after the stomach has been emptied using a wide-bore nasogastric tube. Intravenous correction of dehydration is undertaken and 80 mmol of potassium may be necessary during the first 24 hours .
- In some patients, PPI drugs heal ulcers, relieve pyloric oedema and overcome the need for surgery .
- Endoscopic balloon dilatation of benign stenoses may be possible in some patients but in others partial gastrectomy is necessary; this is best done after a 7-day period of nasogastric aspiration, which enables the stomach to return to normal size.
- A gastroenterostomy is an alternative operation but, unless this is accompanied by vagotomy, patients will require long-term PPI therapy to prevent stomal ulceration.

Zollinger–Ellison syndrome

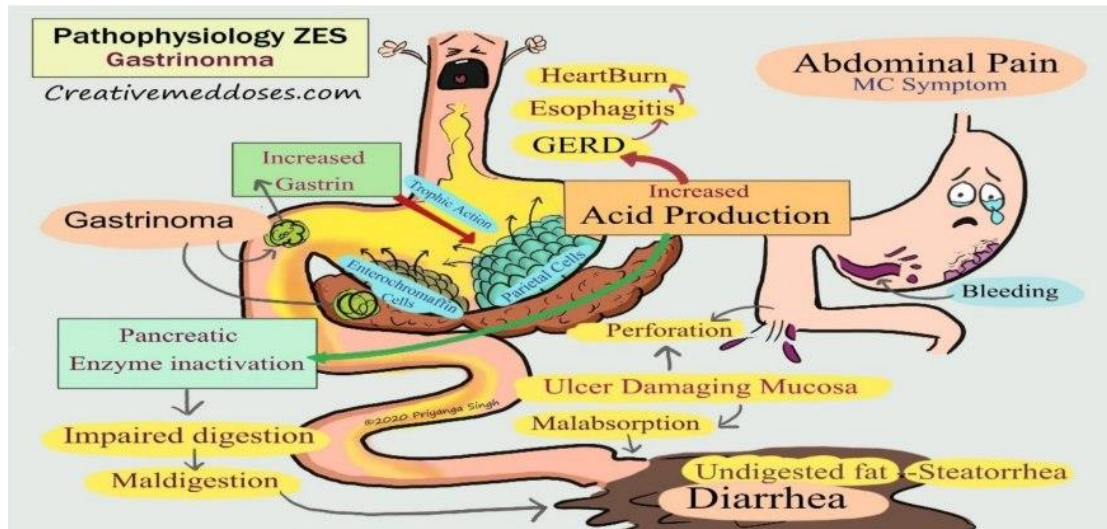
- This is a rare disorder characterised by the triad of severe peptic ulceration ,gastric acid hypersecretion and a neuro-endocrine tumour of the pancreas or duodenum (‘gastrinoma’).
- It probably accounts for about 0.1% of all cases of duodenal ulceration. The syndrome occurs in either sex at any age, although it is most common between 30 and 50 years of age.

Pathophysiology

- The tumour secretes gastrin, which stimulates acid secretion to its maximal capacity and increases the parietal cell mass three- to sixfold. The acid output may be so great that it reaches the upper small intestine, reducing the luminal pH to 2 or less.
- Pancreatic lipase is inactivated and bile acids are precipitated. Diarrhoea and steatorrhoea result .
- Around 90% of tumours occur in the pancreatic head or proximal duodenal wall. At least half are multiple and tumour size can vary from 1 mm to 20 cm .



- Approximately one-half to two-thirds are malignant but are often slow growing .
- Between 20% and 60% of patients have multiple endocrine neoplasia (MEN) type 1.



Clinical features

- The presentation is with severe and often multiple peptic ulcers in unusual sites, such as the post-bulbar duodenum, jejunum or esophagus .
- There is a poor response to standard ulcer therapy .
- The history is usually short, and bleeding and perforations are common .
- Diarrhea is seen in one-third or more of patients and can be the presenting feature.

Investigations

- Hypersecretion of acid under basal conditions, with little increase following pentagastrin, may be confirmed by gastric aspiration .
- Serum gastrin levels are grossly elevated (10- to 1000-fold) .
- Injection of the hormone secretin normally causes no change or a slight decrease in circulating gastrin concentrations, but in Zollinger–Ellison syndrome it produces a paradoxical and dramatic increase in gastrin.
- Tumour localisation (and staging) is best achieved by a combination of CT and EUS; radio-labelled somatostatin receptor scintigraphy.



Management

- Some 30% of small and single tumours can be localised and resected but many tumours are multifocal with metastatic disease and, in these circumstances, surgery is inappropriate .
- In the majority of these individuals, continuous therapy with omeprazole or other PPIs can be successful in healing ulcers and alleviating diarrhoea ,although double the normal dose is required.
- The synthetic somatostatin analogue, octreotide, given by subcutaneous injection, reduces gastrin secretion and may be of value .

Functional dyspepsia

- This is defined as chronic dyspepsia in the absence of organic disease.
- Other commonly reported symptoms include early satiety, fullness , bloating and nausea. 'Ulcer-like' and 'dysmotility-type' subgroups are often reported but there is overlap between these and with irritable bowel syndrome.

Pathophysiology

- The cause is poorly understood but probably covers a spectrum of mucosal ,motility and psychiatric disorders.

Clinical features

Patients are usually young (< 40 years) and women are affected twice as commonly as men. Abdominal discomfort is associated with a combination of other 'dyspeptic' symptoms, the most common being nausea, satiety and bloating after meals. Morning symptoms are characteristic and pain or nausea may occur on waking .Direct enquiry may elicit symptoms suggestive of irritable bowel syndrome. Peptic ulcer disease must be considered, while in older people intra-abdominal malignancy is a prime concern. There are no diagnostic signs, apart from inappropriate tenderness on abdominal palpation, perhaps. Symptoms may appear disproportionate to clinical well-being and there is no weight loss. Patients often appear anxious .A drug history should be taken and the possibility of a depressive illness should be considered. Pregnancy should be ruled out in young women before radiological studies are undertaken. Alcohol misuse should be suspected when early-morning nausea and retching are prominent.



Investigations

The history will often suggest the diagnosis. All patients should be checked for H .pylori infection and patients over the age of 55 years should undergo endoscopy

to exclude mucosal disease. While an ultrasound scan may detect gallstones , these are rarely responsible for dyspeptic symptoms.

Management

- The most important elements are explanation and reassurance. Possible psychological factors should be explored and the concept of psychological influences on gut function should be explained. Idiosyncratic and restrictive diets are of little benefit but smaller portions and fat restriction may help.
- Up to 10% of patients benefit from H. pylori eradication therapy and this should be offered to infected individuals. Eradication also removes a major risk factor for peptic ulcers and gastric cancer but at the cost of a small risk of side-effects and worsening symptoms of underlying gastro-oesophageal reflux disease.
- Drug treatment is not especially successful but merits trial .
- Antacids are sometimes helpful. Prokinetic drugs, such as metoclopramide (10mg 3 times daily) or domperidone (10–20 mg 3 times daily), may be given before meals if nausea, vomiting or bloating is prominent . Metoclopramide may induce extrapyramidal side-effects, including tardive dyskinesia in young patients. H₂-receptor antagonist drugs may be tried if night pain or heartburn is troublesome. Low-dose tricyclic agents, such as amitriptyline, are of value in up to two-thirds.
- Symptoms that can be associated with an identifiable cause of stress resolve with appropriate counselling. Some patients have major psychological disorders that result in persistent or recurrent symptoms and need behavioural or other formal psychotherapy.